Run report: accounting v2 study sessions

Study `research/ecbench_accounting_v2_20261023`

Runs started 2026-10-23 08:15 UTC (batch 1: 08:15 to 08:22; amendment A1 and A4 batches after the audits)

Descriptive only: what ran, on what, with what result recorded. Interpretation is in the study's `REPORT.pdf`. Observations only: no speedup or significance statement is made here.

1. What ran

- Command for each session: `ecbench run --wait --allow-busy --cpus auto --quiet --spec SPEC --out sessions/NAME` (scripts `analysis/run.sh`, `run_a1.sh`, `run_a4.sh`). Audit: `ecbench verify --dir D --replay-all --exit-code` (`analysis/audit.sh`), receipts in `audits/`.
- Binary: branch `./target/release/ecbench`, built at head `7ed39a752`, sha256 `e265bbc00a91...`; the main binary `ecbench-base`, sha256 `a59c4014ff52...`, only for the `*-main` sessions. Host capsule: `host.txt` (env class ECBENV3had995aa2df0f, Intel Xeon 2.10 GHz, 4 logical CPUs, 1 NUMA node, virtual machine, Linux 6.18.44).
- Specs and sha256: `SPEC_SHA256.txt`. Committed specs (unchanged): `check-prime-j0-floor-v2`, `smoke-prime-v2`, `smoke-koblitz-v2`, `claim-koblitz-strong-v2-n13`, `-n19`. New specs: `specs/*.json`.
- Workloads and seeds: one algorithm seed per (workload, round) shared by every arm (spec.rs); target seeds are in each spec (`20261023`, `7`, `202610121301`, `202610121901`, `202610123701`).
- Isolation: the specs ask for L2 (counts need none). The host was a shared 4-vCPU VM with other agents building and testing (load average 2 to 4), so most records earned L1; the level counts are in the table below.

2. Runs requested and completed

Every session completed; every record verified; no failure, timeout or interruption. Audit column: SHA-256 of the receipt of the `--replay-all` audit (`audits/NAME.verify.txt`). `j0-frozen-main` is archived (`archive/`; the sessions of the first j = 0 curve, replaced under amendment A4 because that curve is not registered, are under `archives/fp23-first-run/`) because the branch binary's audit rejects it by design (25 problems: "used 6 automorphisms, the curve has 2"); the main binary's audit of it passes. Every other session: audit OK, 0 problems.

session	session id	status	verified of records	binary sha256	isolation level counts	audit receipt
claim-n13	ECBS1h0417f722e88	complete	256 of 256	e265bbc00a91	L0x64 L1x130 L2x62	b11339d68b29
claim-n19	ECBS1h7f68751e557	complete	256 of 256	e265bbc00a91	L0x64 L1x150 L2x42	82049f3d016c
frozen-koblitz-branch	ECBS1h5466ade4b06	complete	160 of 160	e265bbc00a91	L1x160	ff75f66a1635
frozen-koblitz-main	ECBS1hb8266d1bd07	complete	160 of 160	a59c4014ff52	L1x24 L2x136	2a85c3324dfb
frozen-large-prime-branch	ECBS1hf87ec503d14	complete	32 of 32	e265bbc00a91	L1x16 L2x16	772fd2b210ba
frozen-large-prime-main	ECBS1he690ca5be93	complete	32 of 32	a59c4014ff52	L1x4 L2x28	da9bc0fdce2d
frozen-prime-branch	ECBS1h349b75d5e73	complete	240 of 240	e265bbc00a91	L1x240	64609a593829
frozen-prime-main	ECBS1hc0d0d5b26f8	complete	240 of 240	a59c4014ff52	L1x240	7d4f536c6773
j0-frozen-branch	ECBS1h1c5b6e1e11a	complete	72 of 72	e265bbc00a91	L1x72	b90a3b7d7dd4
j0-frozen-main	ECBS1h7b9bf9250cf	complete	72 of 72	a59c4014ff52	L1x72	9bfa168c88fd
j0-v2	ECBS1h0c64bbd1753	complete	144 of 144	e265bbc00a91	L1x144	458ea2607ed8
pairs-koblitz	ECBS1h001cebd5ff1	complete	360 of 360	e265bbc00a91	L1x360	7bd0076d541b
pairs-large-prime	ECBS1h8ce582cbde1	complete	48 of 48	e265bbc00a91	L1x16 L2x32	85ad837c7ac5
pairs-prime	ECBS1h5c19d0d7e48	complete	480 of 480	e265bbc00a91	L1x480	e97bb70352ca
smoke-koblitz-v2	ECBS1h00aa071a40d	complete	48 of 48	e265bbc00a91	L1x40 L2x8	c388521dc72b
smoke-prime-v2	ECBS1hdbf5bc198da	complete	72 of 72	e265bbc00a91	L1x64 L2x8	e1610031ce56
strong-n37	ECBS1h61cd8e1c170	complete	200 of 200	e265bbc00a91	L1x122 L2x78	6a8bb3a3da97

3. Measured values

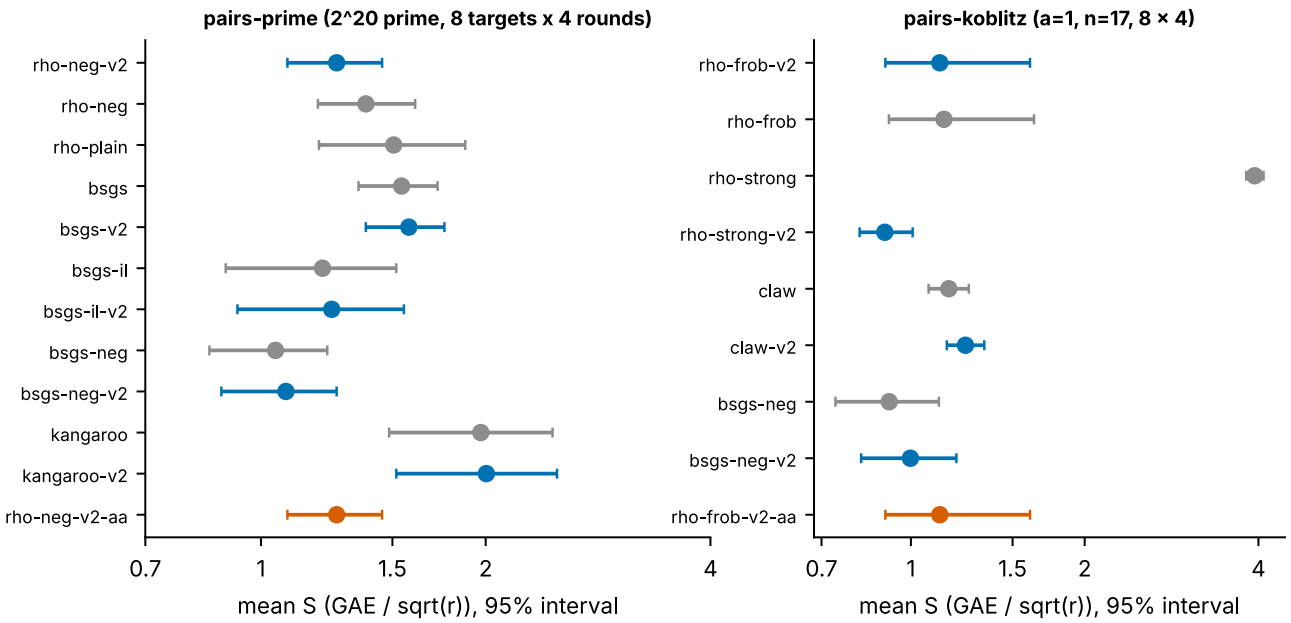


Figure 1: Mean S per arm, `pairs-prime` ECBS1h5c19d0d7e485 and `pairs-koblitz` ECBS1h001cebd5ff12, with the `ecbench table` 95% interval (32 verified runs per arm: 8 targets, 4 rounds). Blue: v2 ids; grey: frozen ids; orange: A/A control. Data: `run_mean_s_per_arm.data.json`.

session	arm	method	verified	mean S	95% interval
pairs-prime	rho-neg-v2-aa	rho.negation_v2	32/32	1.263	[1.085, 1.453]
pairs-prime	kangaroo-v2	kangaroo.vow_v2	32/32	2.004	[1.518, 2.492]
pairs-prime	kangaroo	kangaroo.vow	32/32	1.970	[1.485, 2.458]
pairs-prime	bsgs-neg-v2	bsgs.negation_v2	32/32	1.080	[0.885, 1.263]
pairs-prime	bsgs-neg	bsgs.negation	32/32	1.046	[0.853, 1.227]
pairs-prime	bsgs-il-v2	bsgs.interleaved_v2	32/32	1.244	[0.930, 1.554]
pairs-prime	bsgs-il	bsgs.interleaved	32/32	1.209	[0.897, 1.518]
pairs-prime	bsgs-v2	bsgs.textbook_v2	32/32	1.578	[1.382, 1.761]
pairs-prime	bsgs	bsgs.textbook	32/32	1.543	[1.351, 1.724]
pairs-prime	rho-plain	rho.plain	32/32	1.506	[1.196, 1.878]
pairs-prime	rho-neg	rho.negation	32/32	1.382	[1.192, 1.609]
pairs-prime	rho-neg-v2	rho.negation_v2	32/32	1.263	[1.085, 1.453]
pairs-koblitz	rho-frob-v2-aa	rho.signed_frobenius_v2	32/32	1.122	[0.903, 1.608]
pairs-koblitz	bsgs-neg-v2	bsgs.negation_v2	32/32	0.998	[0.820, 1.199]
pairs-koblitz	bsgs-neg	bsgs.negation	32/32	0.917	[0.740, 1.118]
pairs-koblitz	claw-v2	claw.pair_table_v2	32/32	1.243	[1.154, 1.340]
pairs-koblitz	claw	claw.pair_table	32/32	1.163	[1.073, 1.260]
pairs-koblitz	rho-strong-v2	rho.signed_frobenius_strong_v2	32/32	0.901	[0.815, 1.007]
pairs-koblitz	rho-strong	rho.signed_frobenius_strong	32/32	3.941	[3.809, 4.082]
pairs-koblitz	rho-frob	rho.signed_frobenius	32/32	1.141	[0.916, 1.634]
pairs-koblitz	rho-frob-v2	rho.signed_frobenius_v2	32/32	1.122	[0.903, 1.608]

Mean S is in GAE per $\sqrt{r}$ (`ecbench table`). The interval is the harness's recorded one; where two arms' intervals overlap it says nothing about their difference, which only the paired `ecbench compare` outputs (`analysis/compare/`) address. All S values here carry the harness's `lower_bound` flag where an arm leaves counted work uncharged.

4. References and controls

The reference arms are `rho.negation_v2` (prime) and `rho.signed_frobenius_v2` (Koblitz); every `pairs-*` and `strong-n37` session has an A/A repeat of its reference, which equals the reference on every count (paired ratio 1.000 [1.000, 1.000]). Wall time was recorded but not used: solves are 0.1 to 1.5 ms, below the harness's 1 ms admission minimum, at L1.

5. Raw records

`sessions/NAME/records.jsonl`, `session.json`, `plan.json`, `host.json`, `spec.json` for each session above; `analysis/summary/NAME.json` (jq over the records), `analysis/table/NAME.md`, `analysis/band/NAME.txt`, `logs/run.log`, `logs/run_a1.log`.